

Statistical models

CFTR AlphaGenome splice benchmark — pre-registration supplementary file

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Overview

The study evaluates three fixed in silico splice predictors on a frozen set of CFTR variants. No regression coefficients are estimated; no covariates are adjusted for; no mixed or hierarchical model is fitted. The confirmatory analyses are (i) paired discrimination comparisons using ROC-AUC and DeLong inference, and (ii) a descriptive rank-correlation on the smaller quantitative stratum. All statistics are pre-declared. No omnibus test, no follow-up contrasts, no interactions.

1. Primary confirmatory analysis (Stratum 2, n = 52)

For each variant v in Stratum 2 ($n = 52$; 43 splice-affecting, 9 no-effect; Joynt 2020), let $y(v) \in \{0, 1\}$ be the binary ground-truth label and $S_M(v)$ the per-variant score from predictor M , for $M \in \{\text{AlphaGenome}, \text{SpliceAI}, \text{Pangolin}\}$, computed as specified in the Indices supplementary file.

Empirical ROC-AUC

The AUC for predictor M is the Mann-Whitney U form:

$$\widehat{\text{AUC}}_M = \frac{1}{n_1 n_0} \sum_{i: y_i = 1} \sum_{j: y_j = 0} \Psi(S_M(v_i), S_M(v_j))$$

with $n_1 = 43$, $n_0 = 9$, and $\Psi(a, b) = 1$ if $a > b$, 0.5 if $a = b$, 0 if $a < b$ (ties handled by the standard mid-rank convention).

Paired DeLong test (H1: AlphaGenome vs SpliceAI)

The paired AUC difference is:

$$\widehat{D}_{\text{AG, SAI}} = \widehat{\text{AUC}}_{\text{AG}} - \widehat{\text{AUC}}_{\text{SAI}}$$

Variance is estimated by the DeLong et al. (1988) covariance form, which accounts for the fact that both AUCs are computed on the same variants:

$$\widehat{\text{Var}}(\widehat{D}_{\text{AG, SAI}}) = \widehat{\text{Var}}(\widehat{\text{AUC}}_{\text{AG}}) + \widehat{\text{Var}}(\widehat{\text{AUC}}_{\text{SAI}}) - 2 \widehat{\text{Cov}}(\widehat{\text{AUC}}_{\text{AG}}, \widehat{\text{AUC}}_{\text{SAI}})$$

The test statistic is $Z = D / \sqrt{\text{Var}(D)}$, referred to a standard normal distribution; the 95% CI is $D \pm 1.96 \cdot \sqrt{\text{Var}(D)}$. Two-sided significance is retained (a directional H1 is stated but the test is two-sided so a strong negative result — SpliceAI meaningfully better — is also publishable evidence against H1).

Pass criterion: AUC point estimate for AlphaGenome exceeds that of SpliceAI **and** the two-sided paired DeLong p-value < 0.05 .

Reject: AlphaGenome AUC $\leq$ SpliceAI AUC, or $p \geq 0.05$.

Paired non-inferiority (H2: AlphaGenome vs Pangolin)

The identical DeLong machinery is applied to the AlphaGenome vs Pangolin pair. H2 is a non-inferiority hypothesis with a pre-specified margin $\delta = 0.05$ on the AUC scale:

$$H_2^0: \text{AUC}_{\text{AG}} - \text{AUC}_{\text{PAN}} \leq -\delta \quad H_2^1: \text{AUC}_{\text{AG}} - \text{AUC}_{\text{PAN}} > -\delta$$

Pass criterion: lower bound of the paired DeLong 95% CI on $(\text{AUC}_{\text{AG}} - \text{AUC}_{\text{PAN}}) > -0.05$.

Reject: lower bound of the 95% CI ≤ -0.05 .

Multiple-comparison control

Two families of tests are declared — H1 (superiority vs SpliceAI) and H2 (non-inferiority vs Pangolin) — each with its own pre-specified $\alpha = 0.05$. Because the two hypotheses answer distinct scientific questions on the same paired design, no gatekeeping or Bonferroni correction is applied; each hypothesis is tested at its own two-sided 0.05 level and both outcomes are reported.

Software implementation

Primary implementation: R package pROC (Robin et al. 2011), `roc.test(method = "deLong", paired = TRUE)`. Cross-check implementation: Python `scikit-learn.metrics.roc_auc_score` for the AUC point estimates plus a manual paired-DeLong routine ported from Sun and Xu (2014). Both implementations are run; disagreement > 0.005 in any AUC or > 0.01 in the p-value triggers a manual investigation and is disclosed in the results.

2. Secondary descriptive analysis (Stratum 1, $n \approx 9$)

For each variant v in Stratum 1 ($n = 8\text{--}9$ unique variants, up to 14 data points with multi-organ replicates retained), let $x(v) \in [0, 100]$ be the published percentage residual normally spliced transcript and $S_M(v)$ the per-variant score from predictor M .

Spearman rank correlation

For each predictor M :

$$\hat{\rho}_M = 1 - \frac{6 \sum_{i=1}^n (\text{rank}(x_i) - \text{rank}(S_M(v_i)))^2}{n(n^2 - 1)}$$

reported with a 10,000-iteration bootstrap 95% percentile confidence interval (paired resampling: each bootstrap sample draws variants with replacement and retains the paired $(x, S_{\text{AG}}, S_{\text{SAI}}, S_{\text{PAN}})$ tuple).

No inferential test. Given $n \approx 9$, no p-value is reported and no pass/reject decision is made from this stratum. The correlations are descriptive only.

Multi-organ replicate handling

For c.3717+12191C>T (five multi-organ measurements from Chiba-Falek 1999), Spearman ρ is reported both ways — collapsed to a per-variant mean and with all five points retained. If $|\rho_{\text{collapsed}} - \rho_{\text{retained}}| \geq 0.2$, the collapsed version is treated as primary and the divergence is disclosed in the results.

3. Exploratory analyses (not confirmatory)

The following are declared as exploratory and are separated from the confirmatory results in the manuscript.

Stratum 3 (CFTR2 non-canonical splice regulators). Up to ~19 adjudicated variants from the CFTR2 database. Same three predictor scores computed. Reported as AUC point estimates with DeLong 95% CIs; no formal hypothesis test.

Data-leakage stratification (Stratum 2). Report AUC point estimates with DeLong 95% CIs separately for (i) variants whose Stratum 2 publication post-dates the AlphaGenome training cutoff and (ii) variants published before that cutoff. Descriptive comparison of AUCs across the two sub-strata is reported; no formal interaction test is performed.

Expressivity tally. Descriptive count of which predictors emit a continuous residual-usage estimate vs a binary spliceogenicity call. No statistical test.

4. Model assumptions and controls

Assumption checks. The DeLong procedure is nonparametric on the AUC scale and requires only independent variants and matched (paired) predictor outputs. Independence is assumed at the variant level within each stratum (pre-registered non-independence: five multi-organ points for c.3717+12191C>T on Stratum 1, handled as above; excluded from the Stratum 2 primary test).

The DeLong 95% CI approximation degrades in small samples with extreme AUCs; as a robustness check on the primary test, a paired 10,000-iteration stratified bootstrap 95% CI on the AUC difference is computed (resampling cases and controls independently, retaining variant-level pairing across predictors) and reported alongside the DeLong CI. If the two CIs give qualitatively different conclusions with respect to the pass/reject criteria, the bootstrap CI is treated as primary and this is disclosed.

Positive controls. AUCs are also reported for the raw canonical splice site distance (a trivial nucleotide-distance feature) as a floor benchmark and for a per-variant random score (mean AUC 0.5) as a null benchmark. Any predictor whose AUC lower CI does not exceed the distance-benchmark upper CI is flagged.

Negative controls. Predictor scores are permuted 1,000 times across variants (variant labels shuffled) to construct a null distribution of AUCs per predictor. The observed AUC is required to exceed the 95th percentile of the permutation distribution; failure is disclosed but does not automatically block reporting.

Convergence. DeLong is closed-form; bootstrap and permutation use fixed seeds (documented in the analysis code) so runs are exactly reproducible.

Missing predictions. Any variant × predictor cell with no returned score is excluded pairwise from the affected paired test; counts and reasons reported. If missingness exceeds 10% for any predictor on Stratum 2, the corresponding paired test is downgraded to exploratory and disclosed as such.

5. Contingent decisions

DeLong vs bootstrap disagreement. If the paired DeLong CI and the paired stratified bootstrap CI on the AUC difference lead to different pass/reject decisions under the pre-specified criteria, the bootstrap CI is primary.

Implementation cross-check disagreement. If the R pROC implementation and the Python cross-check diverge by > 0.005 in any AUC or > 0.01 in the DeLong p-value, the discrepancy is investigated manually and disclosed before any pass/reject decision is finalised.

Missing-prediction excess. If missing predictions exceed 10% for any predictor on Stratum 2, the paired test involving that predictor is downgraded to exploratory.

Sample-size shortfall. If a post-freeze amendment causes the Stratum 2 sample size to fall below $n = 30$, the primary confirmatory analysis is downgraded to a feasibility census (descriptive AUCs and CIs only, no pass/reject decision).

Licence-scope contingency. If a DeepMind licence clarification received before submission indicates that rank-order AUC computation on AlphaGenome outputs falls outside permitted non-commercial research use, all AlphaGenome AUCs and paired tests are withheld from publication and only the expressivity tally is reported.

References

DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*. 1988;44(3):837–845.

Robin X, Turck N, Hainard A, et al. pROC: an open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics*. 2011;12:77.

Sun X, Xu W. Fast implementation of DeLong's algorithm for comparing the areas under correlated receiver operating characteristic curves. *IEEE Signal Processing Letters*. 2014;21(11):1389–1393.